

PERCENTAGES





Lesson Outcome

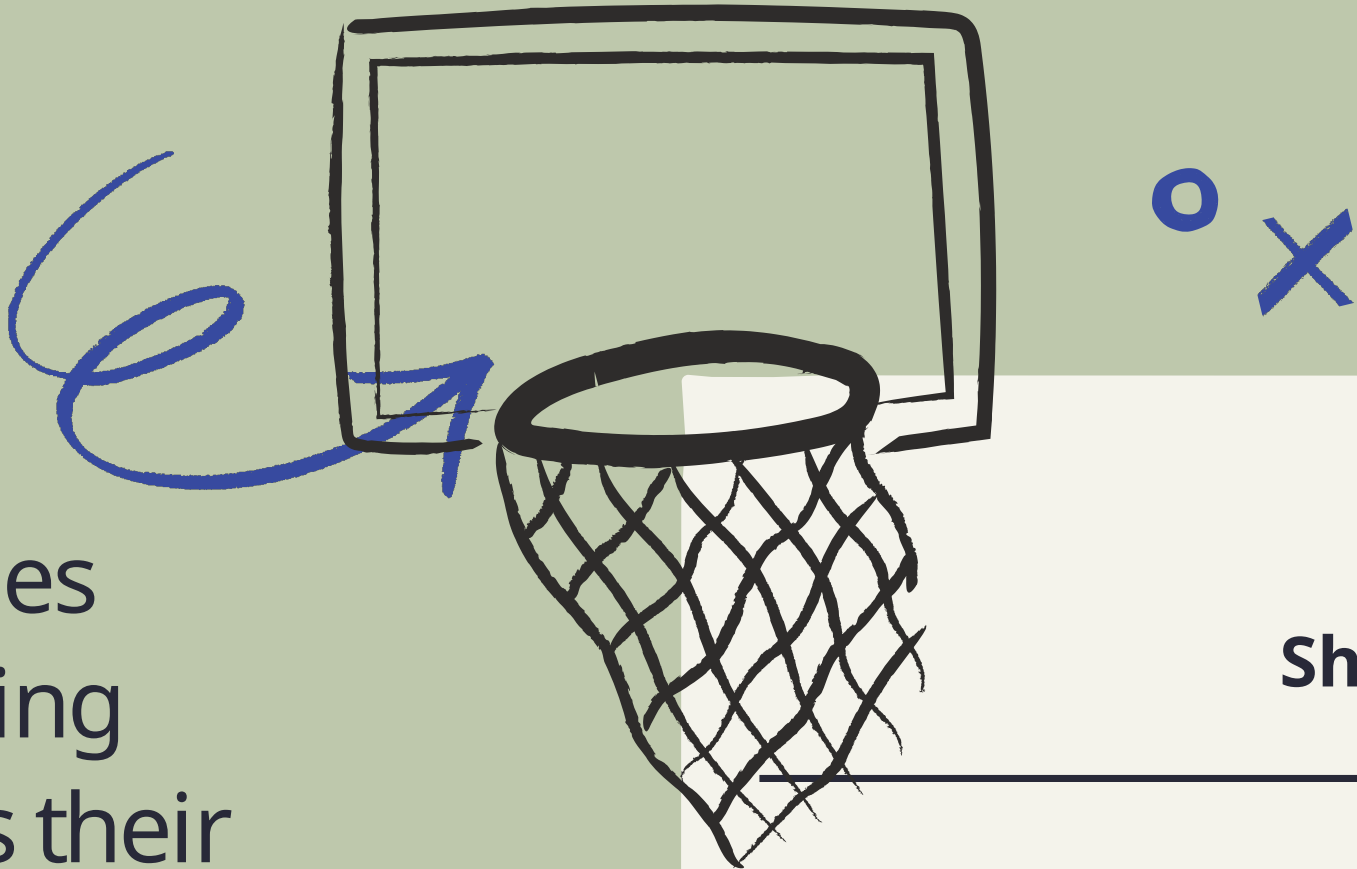
Understand the Concept of Percentage

Convert Between Fractions, Decimals, and Percentages

Solve Real-World Problems Involving Percentages

Michael, Bill and Charles participated in a shooting game. The table shows their respective scores.

Who's the best shooter?



	Shots Made	Attempts
Michael	7	10
Bill	13	20
Charles	15	25



Who's the most accurate shooter?

COMPARING PERCENTAGES



	Shots Made	Attempts	Percentages
✦ Michael	7	10	70%
Bill	13	20	65%
Charles	15	25	60%

Michael is the best shooter, with the highest shot percentage out of all the attempts made!

Find the following



1) 37.5% of 976 =

a. 324

b. 388

c. 366

d. 372

2) 62.5% of 704 =

a. 440

b. 430

c. 460

d. 470

3) $11 \frac{1}{9}\%$ of 918

a. 94

b. 96

c. 114

d. 102



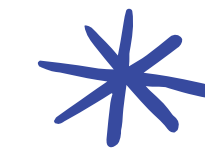


How do we express percentages into fractions?

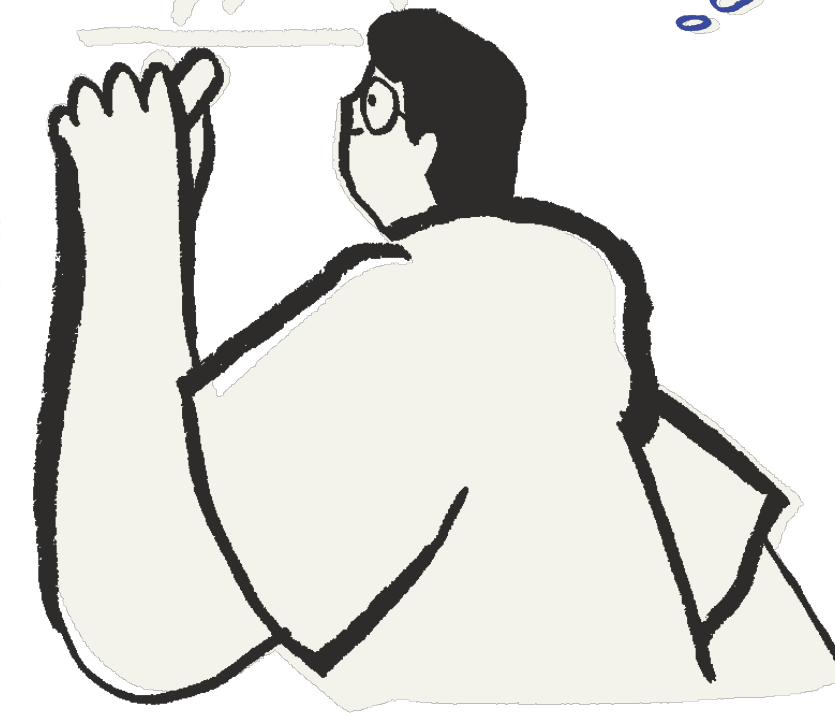
% PERCENT MEANS OUT OF 100

Percentages to Fractions

Percentage	Fraction	Percentage	Fraction	Percentage	Fraction
100%	1	$9\frac{1}{11}\%$	$\frac{1}{11}$	$66\frac{2}{3}\%$	$\frac{2}{3}$
50%	$\frac{1}{2}$	$8\frac{1}{3}\%$	$\frac{1}{12}$	75%	$\frac{3}{4}$
$33\frac{1}{3}\%$	$\frac{1}{3}$	$7\frac{9}{13}\%$	$\frac{1}{13}$	40%	$\frac{2}{5}$
25%	$\frac{1}{4}$	$7\frac{1}{7}\%$	$\frac{1}{14}$	60%	$\frac{3}{5}$
20%	$\frac{1}{5}$	$6\frac{2}{3}\%$	$\frac{1}{15}$	80%	$\frac{4}{5}$
$16\frac{2}{3}\%$	$\frac{1}{6}$	6.25%	$\frac{1}{16}$	$83\frac{1}{3}\%$	$\frac{5}{6}$
$14\frac{2}{7}\%$	$\frac{1}{7}$	$5\frac{15}{17}\%$	$\frac{1}{17}$	37.5%	$\frac{3}{8}$
12.5%	$\frac{1}{8}$	$5\frac{5}{9}\%$	$\frac{1}{18}$	62.5%	$\frac{5}{8}$
$11\frac{1}{9}\%$	$\frac{1}{9}$	$5\frac{5}{19}\%$	$\frac{1}{19}$	87.5%	$\frac{7}{8}$
10%	$\frac{1}{10}$	5%	$\frac{1}{20}$	4%	$\frac{1}{25}$



$$\begin{array}{r} x + 7 = 9 \\ \underline{-7} \\ x = -2 \end{array}$$



MM



4) 64 is what percent of 80?

a. 80%

b. 70%

c. 60%

d. 64%

5) $3/20$ is what percent of $12/25$?

a. 42.75%

b. 31.25%

c. 35.5%

d. 40.5%

6) What percent of 125 is 16?

a. 12.8%

b. 15%

c. 20%

d. 25%



In an examination A got 80 marks and B got 60 marks, then

7) what percent of A 's mark is B 's mark?

a. 80% b. 75%

c. 60% d. 65%

8) By how much percentage A 's mark more than that of B ?

a. 35% b. $33\frac{1}{3}\%$

c. 25% d. $44\frac{4}{9}\%$

9) By how much percentage B 's mark less than that of A ?

a. 35% b. $33\frac{1}{3}\%$

c. 25% d. $44\frac{4}{9}\%$



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Percentage Change

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 $\% \text{ Change} = [\text{Change in Value} / \text{Base Value}] \times 100$ ★★

(1) When the new amount is greater than the original amount, then it is a percent increase.

(2) When the original amount is greater than the new amount, then it is a percent decrease.



Exercise (i)

1) Amir saves 32% of his monthly salary. If he spends Rs. 27200, then find his savings?

- a. 15400 b. 14300
c. 13200 d. 12800

2) Rama's monthly income increased by 25%. Her new monthly income is ₹ 6,000. Find her initial monthly income.

- a. 2400 b. 3200
c. 4800 d. 5600



3) If P gets 25% more than Q , then by what percent does Q get less than P ?

- a. $33\frac{1}{3}\%$ b. 30%
c. $16\frac{2}{3}\%$ d. 20%



4) Venkat's salary was first increased by 20%, then it was decreased by 20%. If his latest salary is ₹ 7200, then what was the initial salary of Venkat?

- a. 7200 b. 7500
c. 6800 d. None of these

Successive Percentage Change

Suppose a number N undergoes a percentage increase of $X\%$ and then $Y\%$, the net change will be

 $\% \text{ Increase} = X + Y + (XY/100)$

Suppose a number N undergoes a percentage increase of $X\%$ and then decrease of $Y\%$, the net change will be

 $\% \text{ Change} = X - Y - (XY/100)$



Suppose a number N undergoes a percentage increase of $X\%$ and then a percentage decrease $X\%$, the net change will be



$$\% \text{ Decrease} = x^2/100$$



5) Anurag gets 25% of the total marks in an examination and fails by 25 marks. Naveen gets 50% of the total marks and gets 20 marks more than the pass mark. Find the total marks.

a. 200

b. 180

c. 300

d. 150

6) In an election there are only three contestants P, Q and R. P secured 25% of the votes and Q secured 60% of the remaining votes. If R secured 9600 votes, by how many more votes has the winner won the election?

a. 3200

b. 4800

c. 5200

d. None of these



7) Keerthana decided to donate 15% of her salary. On the day of donation she changed her decision and donated ₹ 3,600 which is 20% less than what she decided to donate. How much is Keerthana's salary?

- a. ₹ 45,000 b. ₹ 30,000
c. ₹ 28,000 d. ₹ 25,000

8) In an office, totally there are 6400 employees and 65% of the total employees are males. 25% of the males in the office are at least 50 years old. Find the number of males aged below 50 years.

- a. 1040 b. 2080
c. 3120 d. 4160

